

Polynomial Regression Assignment Report

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1. Summary

Two polynomial regression models were trained for the assigned datasets. The final model for var1 used a degree-5 polynomial expansion with ElasticNet regularization, while the final model for var2 used a degree-10 polynomial expansion with ElasticNet regularization.

Problem	Degree	Final model	10-fold CV MSE	10-fold CV R2
var1: Steam turbine optimization	5	ElasticNet, alpha=0.009103, l1_ratio=1.0	0.297031 +/- 0.023987	0.971316 +/- 0.004937
var2: Thermal reservoir mapping	10	ElasticNet, alpha=0.000655, l1_ratio=0.1	0.230232 +/- 0.041747	0.994225 +/- 0.002097

ElasticNet Objective

ElasticNet was fitted on the polynomial feature matrix. The objective minimized was:

$$\text{MSE}(y, X_{\text{poly}} \beta) + \alpha * [\text{l1_ratio} * ||\beta||_1 + (1 - \text{l1_ratio}) * ||\beta||_2^2 / 2]$$

Here, alpha controls overall regularization strength and l1_ratio controls the balance between L1 sparsity and L2 shrinkage.

2. Preprocessing

The var1 dataset has six input variables and var2 has three input variables. Inputs were already normalized to approximately [-1, 1], but scaling was still applied after polynomial expansion because high-degree terms can have different variances.

The pipeline generated polynomial features using PolynomialFeatures(include_bias=False), standardized them with StandardScaler, and fitted a regularized linear model. Keeping these steps inside a pipeline ensured fold-wise scaling during cross-validation and avoided leakage.

3. Model Selection Approach

Ridge regression with alpha=1.0 was used as a stable proxy during degree selection. After selecting the degree, ElasticNet was tuned over alpha and l1_ratio, combining L2 stability with optional L1 sparsity.

4. Data Exploration

Exploratory analysis checked feature ranges, target spread, and whether linear relationships were sufficient. var2 has a wider target spread than var1, while var1 has weak individual feature-target correlations, supporting polynomial expansion.

Problem	Target mean	Target std	Target min	Target max	Largest linear correlations with y
var1	0.8736	3.2708	-10.7271	11.3392	x5: -0.255, x6: -0.227
var2	2.0807	6.6172	-29.9015	32.3655	x2: 0.457, x3: 0.190

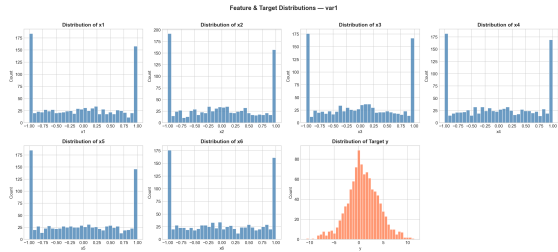


Figure 1a: var1 feature and target distributions.

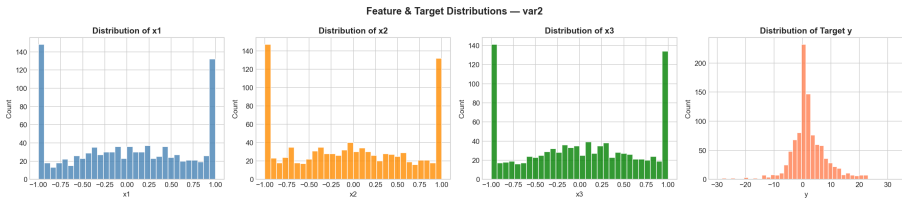


Figure 1b: var2 feature and target distributions.

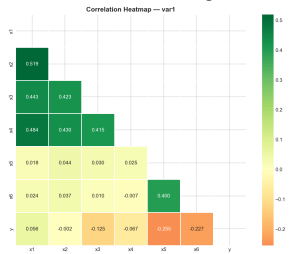


Figure 2a: var1 correlation heatmap.

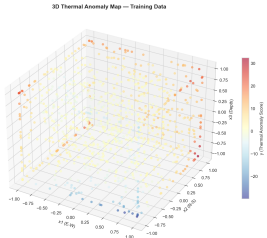


Figure 2b: var2 3D training scatter colored by target value.

5. Polynomial Degree Selection

var1: Degrees 1 to 10 were evaluated. Validation error decreased up to degree 5 and then increased for higher degrees, indicating overfitting. Degree 5 was selected.

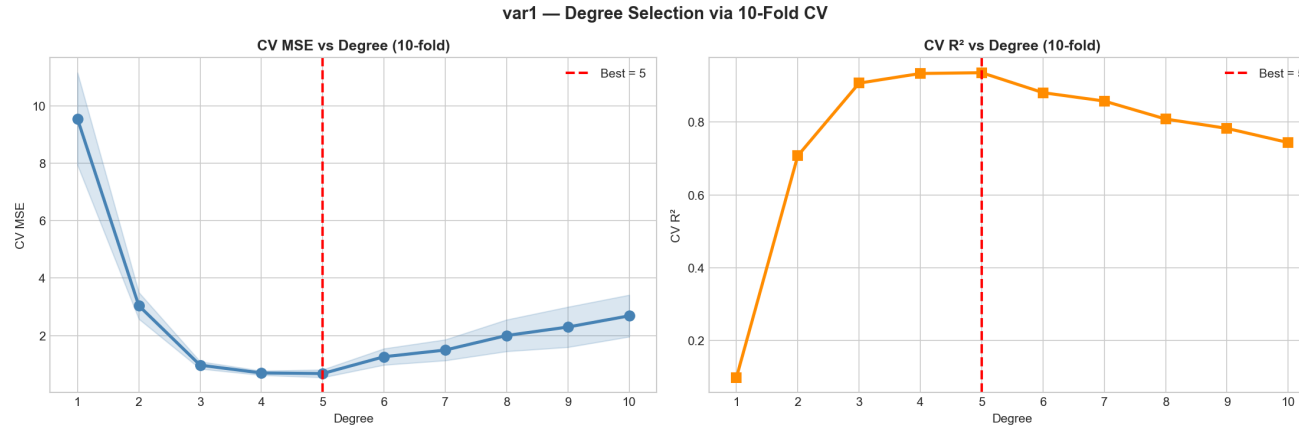


Figure 3: Degree selection for var1 using 10-fold cross-validation.

var2: Degrees 1 to 20 were evaluated. Performance improved until degree 10, after which it plateaued or slightly declined. Degree 10 was selected as the best trade-off between flexibility and generalization.

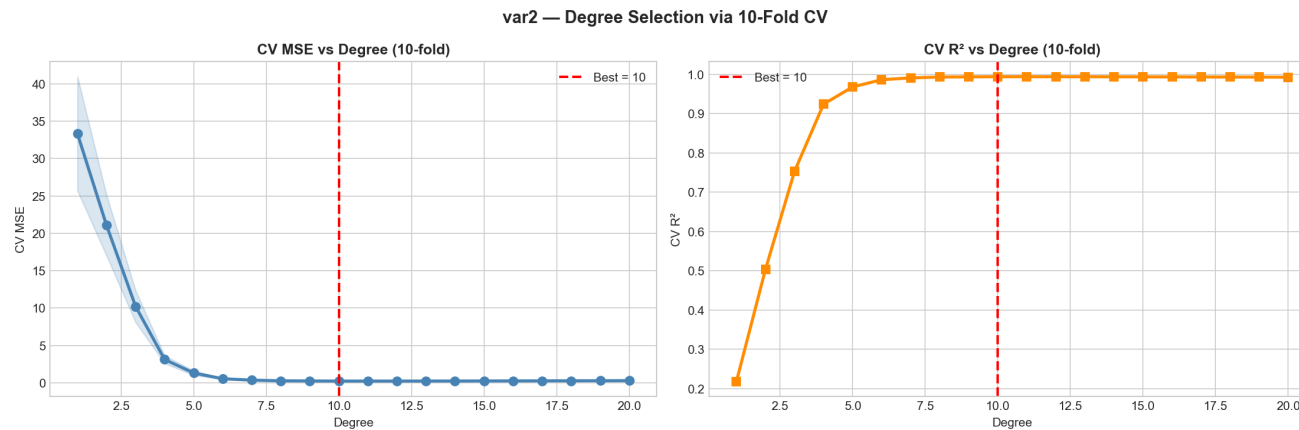


Figure 4: Degree selection for var2 using 10-fold cross-validation.

6. Final Model Details

var1 quantity	Value
Degree	5
Polynomial features	461
Best alpha	0.009103
Best l1_ratio	1.0
Non-zero coefs	126 / 461
Train MSE / R2	0.216335 / 0.979757
CV MSE	0.297031 +/- 0.023987
CV R2	0.971316 +/- 0.004937

var2 quantity	Value
Degree	10
Polynomial features	285
Best alpha	0.000655
Best l1_ratio	0.1
Non-zero coefs	253 / 285
Train MSE / R2	0.162370 / 0.996288
CV MSE	0.230232 +/- 0.041747
CV R2	0.994225 +/- 0.002097

For var1, l1_ratio=1.0 made the model Lasso-like and removed many weak degree-5 terms. For var2, l1_ratio=0.1 kept the model mostly Ridge-like, suitable for a response distributed across many degree-10 terms.

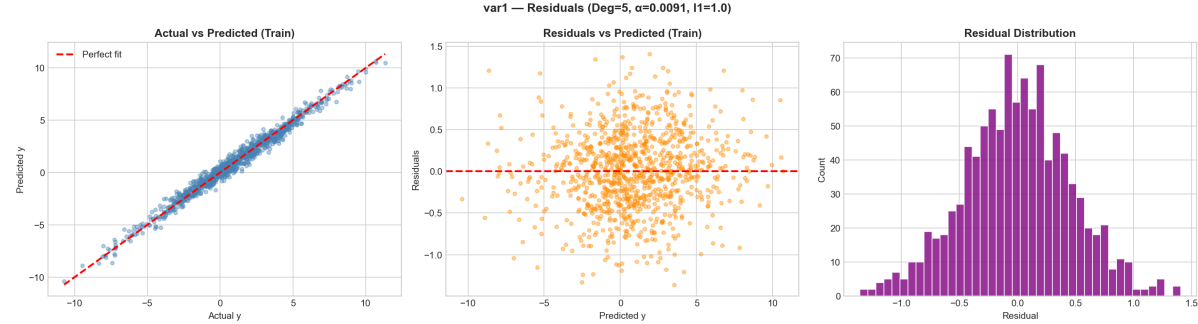


Figure 5a: var1 training residual diagnostics.

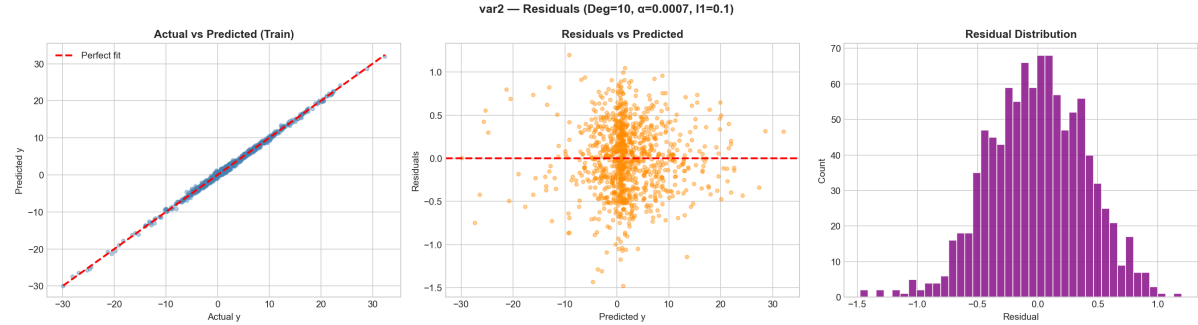


Figure 5b: var2 training residual diagnostics.

7. Prediction and Regularization Diagnostics

Prediction distributions were compared against training target distributions as a sanity check that the final polynomial models produced realistic ranges.

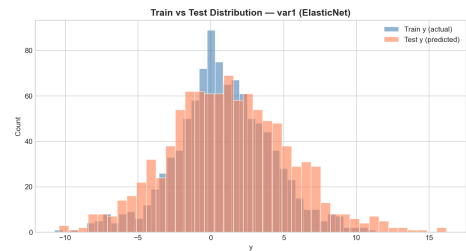


Figure 6a: var1 training target vs test prediction distribution.

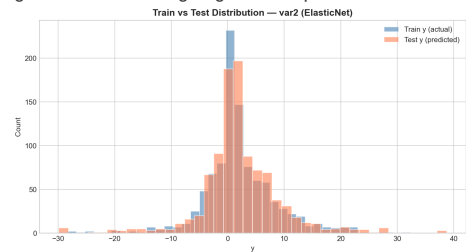


Figure 6b: var2 training target vs test prediction distribution.

Coefficient sparsity explains the regularization behavior: var1 is sparse due to the Lasso-like penalty, while var2 keeps most polynomial coefficients active.

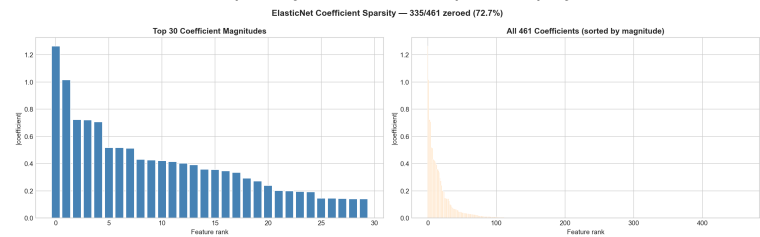


Figure 7a: var1 coefficient sparsity.

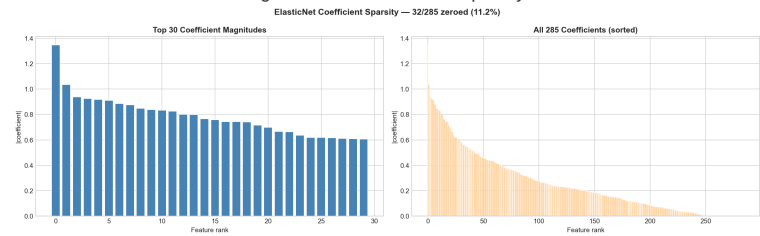


Figure 7b: var2 coefficient sparsity.

8. Conclusion

Degree 5 was selected for var1 and degree 10 for var2 using cross-validation. Final 10-fold CV R2 scores were 0.971316 for var1 and 0.994225 for var2, and predictions were generated from the final fitted pipelines.